

13.1 Social implications: Email and internet safety

In 2015, a computer security firm called Kaspersky Inc. uncovered a computer attack by a group of Russian hackers against banks from around the world. These hackers used a “phishing attack” to infect computers on the bank’s internal network with a virus. Once they gained access to a bank employee’s computer, they used the computer to gain access to the security cameras inside the bank’s offices.



Figure 13.1: *Cybercrime*

The following table shows the common email and internet threats, as well as what you can do to protect yourself from potential threats.

Table 13.1: *The most common threats*

TYPE	DESCRIPTION	PROTECTION
Virus	A computer <i>virus</i> is any program that infects a user’s computer and acts in a way that will harm users without their knowledge. Users do not install viruses; viruses spread automatically, without the users knowing about them.	Keep software up to date, especially your antivirus software. Do not install or run any programs you do not know or trust. Do not download executable email attachments. Run a virus scan on any flash disk that you do not know, before using it.
Trojans	A <i>Trojan horse</i> is another type of virus that pretends to be a useful piece of software. However, once you install it, the Trojan horse infects the computer and can damage it in many ways. One of the most dangerous types of trojan horses is called a keylogger , which records every key you press on the computer and sends the information to a hacker on the internet. This allows the hacker to easily figure out your internet usernames and passwords.	The easiest way to protect yourself from Trojan horses would be to NOT open any emails, or download any attachments from unknown sources. Delete these messages to take care of this problem. Also, by installing antivirus software, this will scan every file that you download to protect you from these types of threats.

... continued

TYPE	DESCRIPTION	PROTECTION
Worms	A <i>worm</i> is a type of virus that can travel from one computer to another computer, without any human interaction. Worms make duplicates of themselves and can send thousands of copies to computers on the same network. This allows worms to quickly spread over networks, making them very difficult to remove!	To prevent worms, you should not click on any links on emails from unknown sources, or even instant messages. Avoid downloading attachments from emails and instant messages from unknown sources. Also, install antivirus software, with a software firewall so that it can block any potential threats from the network.
Hoax	<i>Hoax warnings</i> are used by people as “scare alerts”. Hoax warnings are passed on by users who think that they are helping other people by spreading the warning. The most common virus is the “email hoax”. The email will probably notify the recipient that there is a new destructive virus and that they should pass on this email to everyone in their address book. This results in thousands of unwanted emails that overload the system.	To protect yourself from hoaxes, do not open any emails, or download any attachments from unknown sources. Delete these messages to take care of this problem. Also, by installing antivirus software, this will scan every file that you download to protect you from this type of threat.
Phishing and email spoofing	<i>Phishing and email spoofing</i> attacks try to obtain sensitive information (such as usernames, passwords and banking details) by sending emails to users that look like official emails. These emails will either directly request the sensitive information, or redirect users to an official-looking website from where their information will be stolen.	Never send sensitive information, such as usernames and passwords over email. No company will ever ask for this information via email or phone. Make sure the email address and domain name are correct for any email that looks suspicious.
Pharming	Much like phishing, <i>pharming</i> attacks create an official-looking website that requests sensitive information. A very common pharming attack allows users to “change” their passwords. Instead of changing their passwords, the user’s username and password are recorded and their account is taken over.	Do not follow any links to change your password, unless you have requested the password change. Verify the domain name and web address of any website that requests sensitive information.
Ransomware	<i>Ransomware</i> is a particularly malicious virus that locks all of a user’s files. Thanks to the strong encryption, it can be physically impossible to unlock these files on your own. The only way to gain access to the file is by paying the virus creator a specific amount of money (usually around R3 000) within 72 hours.	Ransomware is simply a virus; so the same advice applies to ransomware. If your computer is infected with ransomware, do research online to find out what happened to other users who either paid, or who did not pay the ransom and then decide if your files are worth the amount being asked.



UNDERSTANDING SPAM AND PHISHING

Have a look at this video to understand more about spam and phishing:



<https://youtu.be/Ni37Jl7KnSc>



Something to know

Computer crimes are also referred to as “cyber crimes” or “e-crimes”.

COMPUTER CRIMES

Although ICTs are useful, various computer crimes can take place if you are not careful enough. In this section, we will look at the following different types of computer crimes:

- Information theft
- Data theft
- Identity theft

INFORMATION THEFT

Information is one of the most valuable resources available to people who are looking to exploit or cause others harm. These kinds of people commit fraudulent acts and can access personal information through hacking into someone’s computer files. We call them *cybercriminals*.

A cybercriminal can have access to the following types of information:

- **Personal information:** Personal information, such as your identity number, can be used to steal your identity. This allows the cybercriminal to open store accounts and apply for loans using your personal information.
- **Financial information:** Information about your banking details can give the cybercriminal access to your banking accounts, steal your money and do online purchasing.
- **Social media information:** If a cybercriminal has access to your social media accounts or information, the cybercriminal can exploit your friends and family by impersonating you. This can be done by sending a message to people asking them for money, or a message containing a virus that will allow the cybercriminal to exploit more people.
- **Interests:** Information, such as your likes and interests, can be very valuable as it can be sold to companies. The companies then use this information to tailor make advertisements and specials that are specifically aimed at you.

To avoid these problems, do not share your sensitive information unnecessarily and make sure to protect your information using a strong and secure password. While sharing information is part of using the internet, be selective about the websites with which you choose to share your information. If you use your email address for every website you visit, do not be surprised if you start receiving consistent spam emails.

IDENTITY THEFT

Identity theft is when someone else pretends to be you. This is very similar to information theft. Here, the criminal is illegally obtaining information about you. This type of criminal can use information, such as your full name, date of birth, passwords, email address and bank card details to gain access to your bank account or smartphone.

Cybercriminals can use this information to steal money from you, or even sell your information to others.

DATA THEFT

Data theft can refer to two different crimes. We will now take a look at each of these in some more detail.

The first is the theft of data caused due to a third-party gaining access and using your internet connection without your knowledge. To protect yourself against this form of data theft, you can do the following:

- Protect your network by using a username and password.
- Make sure to keep your username and password secret and do not share it with people you do not trust.
- If you suspect that someone has obtained access to your network, change your username and password.

The second type of data theft is *hotlinking*. Hotlinking is when someone takes a file or video from their website and links it to your website. This is problematic, as each time a user clicks on the link to download the file or view the video, your bandwidth will be used.



Activity 13.1

Mrs Abrahams loves shopping online. She surfs the internet and every week, she purchases something new. One day she realised that there were purchases made that she did not make herself.

1. Explain how ICTs have benefited Mrs Abrahams in this context.
2. List one disadvantage of ICTs used in this context.
3. Briefly explain what Mrs Abrahams could possibly be the victim of.
4. Suggest what she can do to remedy the situation. Motivate your answer.

SAFE INTERNET AND EMAIL USE

To avoid most of these problems, computer experts suggest the following rules:

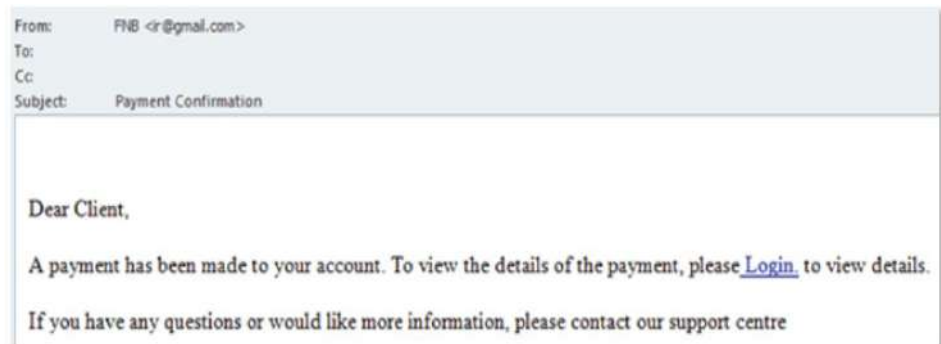
- **Use a secure password:** A good way to create a secure password is to use three or four randomly selected words as your password. Also, include numbers and symbols. “Correct!Horse2Battery&Staple” for example, is easy to remember and difficult to hack.
- **Pay careful attention to the websites and links you open:** Make sure you know where a link is taking you to and verify that the web address is the same as the website you expected to visit.
- **Do not share sensitive information unnecessarily:** While sharing information is part of using the internet, be selective about the websites with which you choose to share your information. If you use your email address for every website you visit, do not be surprised if you start receiving reams of spam emails!
- **Do not open suspicious emails:** You can identify suspicious emails when they are from unknown or suspicious email addresses, emails that seem too good to be true, emails that request sensitive information, emails with executable attachments and emails with poor spelling or grammar.
- **Do not download suspicious programs or attachments:** If you do not know who the sender of the email is, never download the attachment. Even if you know who the sender is, but you are not expecting an attachment, do not open it. Your friend may have been hacked and sending a virus!
- **Keep antivirus applications up to date:** The antivirus application is your last line of defence against viruses. By keeping it up to date, you make sure that it knows how to detect all the latest security threats.

- **Keep other software up to date:** Software weaknesses or vulnerabilities can be used by hackers to gain access to your computer. One of the main reasons that programmers release updates, is to fix these vulnerabilities.
- **Use two-factor authentication whenever possible:** Two-factor authentication prevents anyone from logging into your accounts using just your username and password. Instead, it needs a second factor (which is usually a physical device, such as your phone) to access your account. This means that your account can only be hacked by a small number of people with access to your phone.



Activity 13.2

1. Define the following email terms:
 - a. Hoax
 - b. Spam
2. Study this example of phishing:



- a. Describe two features in the email that confirm the email to be a phishing attempt.
 - b. Explain the main difference between phishing and pharming.
3. Study the following extract from a case and answer the questions that follow:

Mike was the head of a new division of End Corp., a small company with about 45 PCs. Mike was hired to reduce expenses for the company and so he would only authorise the purchase of one copy of each software program that the company uses. His rationale was, "we bought it, and we can do what we want to do with it".

Adapted from: http://www.sjia.net/index.php?option=com_content&view=article&id=338&Itemid=351

- a. Identify the term that describes this extract.
- b. State one possible consequence for the developers of the software programs.
- c. State why the URL was provided in this extract.

REVISION ACTIVITY

1. Write down only the letter matching the correct or best answer.
 - a. A worm is a type of virus that does which of the following?
 - A. Locks your files and demands that you pay money to unlock them
 - B. Spreads itself without human intervention
 - C. Hides your files away
 - D. Pretends to be a legitimate program

(1)

... continued

REVISION ACTIVITY

... continued

2. Bandile is a Standard Bank client. He received the following message from them:

26 February 2019
Good day

Welcome to our digital NEXT, and the old Online Banking site has been discontinued. Upgrade to continue Banking with a two-step sign in feature.

Click Here to sign in and upgrade, you will be required to first enter your email address and then your Password on the next screen.

Note: Registered phone number or Email address we have on our record must be confirmed as we will verify by sending One-Time-PIN to validate your details.

Gmail placed the message in a Spam folder and inserted the following message at the top of the email:



Be careful with this message

Gmail could not verify that it actually came from standardbank.co.za. Avoid clicking links, downloading attachments, or replying with personal information.



- What is this type of email called? (1)
 - Gmail disabled all the hyperlinks in the message. Why was this done? (2)
 - Mention one way that Bandile could recognise that this email was not legitimate. (1)
 - What are spam emails? (1)
- Explain what a hoax email is. Does this type of email normally pose a threat? (2)
 - Explain the difference between data theft and identity theft. (4)
 - You have been asked to create a short presentation for your class on how to stay safe when using the internet and email. Write down five points that you would include in your presentation. (5)

[TOTAL: 17]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

NO.	DID YOU ...	YES	NO
1.	Identify email threats and remedies.		
2.	Understand the different computer crimes.		
3.	Identify ways on how you can protect yourself from being a victim of a computer crime.		
4.	Understand the importance of using email and the internet safely.		

INFORMATION MANAGEMENT

CHAPTER OVERVIEW

- | | |
|-----------|--|
| Unit 14.1 | Concepts used in information management |
| Unit 14.2 | Practical Assessment Task (PAT) overview |
| Unit 14.3 | Sifting data |
| Unit 14.4 | Processing data |
| Unit 14.5 | Presenting data |

At the end of this chapter, you should be able to:

- Understand data versus information.
- Describe the problem-solving steps.
- Explain how to gather data and information from different sources.
- Understand the difference between information and knowledge.
- Understand the importance of questionnaires and surveys.
- Sift out data.
- Process data.
- Present the data in the form of graphs or a report.

INTRODUCTION

We are bombarded with a large amount of information on a daily basis. This information comes in many different forms. In order to gain knowledge and understanding, you need to know how to manage this information.

Information management (IM) is the process of gathering, selecting, processing and then presenting the information to people. In this chapter, we will look at data versus information, and how to understand a problem or task, as well as the importance of questionnaires. Lastly, we will look at different information sources and data-gathering tools.

14.1 Concepts used in information management

The information management process is used in every environment; be it at school or in a business. This process is used to determine if set goals are being achieved, or whether or not a company is making a profit. Managing information can be broken down into the following three steps:

1. Inputs
2. Processing
3. Outputs

Data is an example of input. In Chapter 1, we looked at data and information. You learned that data is raw, unorganised numbers, signals, or facts and by processing this data, it becomes information. Information consists of facts and numbers that have been organised so that they are useful for people.



Case Study

Tracking class performance

For example, if the Mathematics teacher wanted to see how well the current class is performing compared to last year's class, she might ask the school's database to convert its data into averages for the two years. That way, many pages of data are converted into two numbers that can be compared easily. Similarly, the report you receive at the end of each school year takes all the data that the teachers collected throughout the year and turns that data into a single report that you can use to measure your performance.

DATA

sector	message
00nl_Combined_Sector	14625
00nl_Combined_Sector	10125
00nl_Combined_Sector	4500
churches	1350
year	3150
00nl_Combined_Sector	5625
churches	4950
year	675
00nl_Combined_Sector	4500
00nl_Combined_Sector	1090

INFORMATION

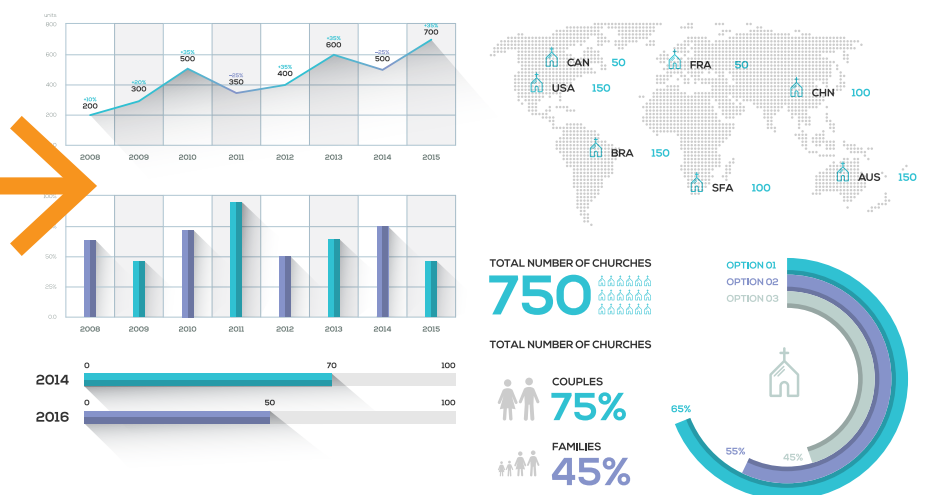


Figure 14.1: Data converted into information



The information management process in the Practical Assessment Task (PAT) can be broken down into the following phases:

- **Phase 1 – Inputs:** This phase is characterised by the gathering of *quality* information. For example, you have to research a specific topic. At this phase, you will have to create questions that identify what problems you will encounter with this topic. This will help you to understand the kind of quality information for which you will need to ask. After identifying the problems, you will need to get the relevant data and information from different sources. This could be in the form of printed media, or searching for information on the internet that you learned about in Chapter 6.
- **Phase 2 – Processing:** After gathering all the data (inputs), you will need to process it. You must understand what the information is and you will have to find solutions to the problems that you identified in Phase 1. For example, if you used a questionnaire, you will need to use a computer application like a spreadsheet to process the data and analyse the information.
- **Phase 3 – Outputs:** The last phase involves presenting that analysed information in the form of a report or presentation. Therefore, you will need to apply what you learned in word processing, spreadsheets and presentations in order to make the information understandable to other people.

14.2 Practical Assessment Task (PAT) overview

Before you can start a research project, such as the PAT, you must understand what the task is about. The purpose of the PAT is to test:

1. Your information management skills
2. Your application skills (to produce quality outputs using what you learned during the year)

Take note that the PAT is an important component in your year's marks. You need to complete the PAT on time; otherwise, it will affect your overall results.

TECHNIQUES USED IN INFORMATION MANAGEMENT

There are different techniques used in the information management process. These techniques include identifying the task definition and information sources.

The PAT is usually broken up into three phases:

- Phase 1 (Term 2): Understanding the task, using information-gathering tools and using questionnaires to obtain data (input).
- Phase 2 (Term 3): Processing the data that you obtained in Phase 1, including using the knowledge you acquired earlier in the year (i.e. using spreadsheets) (processing).
- Phase 3 (Term 4): Presenting the information that you analysed in Phase 2 in the form of presentations or reports (output).

UNDERSTANDING THE TASK

First of all, you need to understand what the topic is about, as well as what the task requires you to do.

It is important to manage your data and information effectively. You should ensure that you have a good filing structure.

CREATING A GOOD FILING STRUCTURE

To make your life easier, create folders and sub-folders to store the data and information according to the three phases. Make sure you name all the folders and sub-folders correctly so that it saves you time and effort.

You can name the main folder using your name and surname, and name the three sub-folders Phase 1, 2 and 3, as shown below:



Figure 14.2: Folder structure

TASK DEFINITION

The *task definition* given by the PAT will not explain the task in much depth. You will be required to interpret, as well as apply the instructions and problem definition given in the PAT.

It is important that you develop your own task definition written in your own words. When developing your own task definition, you can expand on the definition of the task by using mind maps or bulleted lists.

An example of a task definition is shown below:

I am doing this investigation to report to <insert target audience> about <insert PAT topic>.

I will find information on <insert PAT topic>.

I will use the following <source> <information-gathering tools> and create questionnaires to conduct interviews.

I will also use the internet to help me find more information.

After I have received all the data, I will use spreadsheets to process and analyse the data so that it converts to meaningful information.

Using this information, I will then create a report. I will also use the questionnaire answers to create a report using a word processor.

Lastly, I will translate the information into a slide presentation to show my <insert target audience>.

Figure 14.3: An example of a task definition

A PAT project requires you to think and apply your own creativity. When you are given a PAT project and before you can even start with Phase 1, you need to think of the following:

1. Read the PAT task scenario.
2. Write down what your initial thoughts are about the scenario.
3. Jot down the information that is given and required.
4. Identify a focus area that is related to the scenario.
5. Write down the possible sub-topics.

PROBLEM SOLVING

Problem solving is the process of looking at a situation and finding exactly what the problem is, what the causes of this problem are, possible solutions, or how the problem can be removed.

In any problem or scenario where a decision must be made, there are steps that you can follow. The problem-solving process can be broken down into the following five steps:

1. Identify the problem
2. Understand the problem
3. Gather information
4. Find a strategy
5. Implement the strategy

You can ask yourself the following questions when identifying a problem:

1. What is the problem?
2. Is it my problem?
3. It is worth solving?

After defining the problem or task, you need to think about how to take the process further. You need to decide on how you will take on the task. The best way to do this is by finding the information. To get the information, you must ask yourself questions, such as:

- What, when, where and why?
- If, what if and how?

We just spoke about problem solving in general. In Phase 1 of the PAT, you need to identify the task that is given to you and set questions in order to help you investigate the task efficiently.

When setting questions, keep the following in mind:

- Questions can be arranged according to sub-topics.
- You can write down all your questions and then later categorise them according to their sub-topics.
- Questions can be arranged into levels and when sorting out your questions, you can sort them according to the question level.

Table 14.1: *Sorting questions into levels*

LEVEL	TYPE OF QUESTION
Level 1	These are questions that can be answered simply by the facts and starts with words, such as: What? When? Where? Who? How many?
Level 2	These are questions that help you explore or investigate and starts with the word, such as: Why? How?
Level 3	These are questions that will help you to predict or assume and starts with the word: 'If'. For example: <i>What if _____?</i>
Level 4	These are questions that will help you to review, or make a judgement or a conclusion of some sort and usually starts with questions, such as: Would it be better if _____? What recommendation? How can I determine what would be the best way?

INFORMATION SOURCES

Once you have an understanding of the task and have developed the task definition, you will need to gather data and information. In this phase, you must consult several types of information sources to understand the subject better.

Information can be gathered using the following sources:

- **Electronic media:** Web pages, e-books, the internet, online books and encyclopaedias
- **Printed media:** Books, encyclopaedias, journals and magazines
- **People:** Surveys, interviews and questionnaires
- **Other sources:** Social media and your own knowledge

Table 14.2: *The different types of information-gathering sources*

TYPE OF SOURCE	DESCRIPTION	ADVANTAGES	DISADVANTAGES
Electronic media	Data and information can easily be accessed on the internet. This can be done by using online resources, such as Wikipedia, Google searches and internet articles, books, and magazines.	• A wealth of information available • Easily accessible • A variety of resources	• Lots of information to sift through • Time consuming when looking at specific types of information • Not all sources are accurate or reliable • Source can be removed and made no longer available
Printed media	Data can also be gathered from printed media, such as books, newspaper articles, magazines and procedures.	• Information can be stored and is available when needed	• Have to carry media around • Limited amount of information • Can sometimes be outdated
Questionnaires	Questionnaires are a set of written or printed questions, usually with a choice of answers. They can be conducted online, or on paper. Questionnaires are usually constructed for the purpose of a survey or statistical study.	• Reliability of the information can easily be verified • Cost effective • Data is received instantly if it is done online • Answers are usually easy to analyse	• Writing up a questionnaire with the correct questions can be difficult • Sometimes people do not elaborate when answering questions and might be limited to one-word answers • Respondents might not be easily reachable
Surveys	A survey is the process of collecting information from a group of people.	• Can be developed in less time than other data-collection methods • Cost effective • Can be conducted remotely, for example over the internet, on mobile devices, over email, or telephonically • Can collect data from a large number of people • A wide range of data can be collected, for example the attitudes, opinions, beliefs and values that people hold	• People may not necessarily give the most accurate, honest answers • Surveys with closed-ended questions may be less valid than other question types • Questions may be interpreted differently if they are not clear and, therefore, not give accurate answers • “Yes” or “no” answer options can limit the process of analysing the data
Interviews	An interview is generally in the form of a formal meeting where a set of questions are asked face to face, over a software application such as Skype, or telephonically. Interviews can be conducted to find out if someone is suitable for a specific job, or a course of study, for example.	• Information is reliable • Cost effective • Data is received instantly • Credibility of the information can be verified	• Creating an interview with the correct questions can be difficult • Sometimes people do not elaborate when answering questions and might be limited to one-word answers • Analysing responses can be time consuming • Respondents might not be easily reachable

MORE ABOUT INFORMATION-GATHERING SOURCES

In the PAT, you will usually be given the instructions to conduct a survey or questionnaire where you must interview people. They will answer the questions that you set in the questionnaire and you will then collect all the questionnaires and use the information gathered from the questionnaires for an important task when completing the PAT.

CUSTOMER SATISFACTION SURVEY

	Excellent	Good	Average	Poor
1) Overall, how satisfied are you with the service you received?	☒	☐	☐	☐
2) To what extent did the services meet your expectations?	☒	☐	☐	☐
3) How courteous and attentive was our staff?	☒	☐	☐	☐
4) Did you receive services in the time expected?	☐	☐	☐	☐
5) How do we compare against similar businesses?	☐	☐	☐	☐
6) Would you use our service again?	☐	☐	☐	☐
7) How likely is it that you would refer a friend to us?	☐	☐	☐	☐

out us? better? out your contact information here.

Figure 14.3: Example of a questionnaire

The following are tips on how to create questionnaires; whether it is on a Word document, or using an application, such as Google Forms:

- Try using existing questionnaires and adapt the questions to better suit the information for which you are looking
- Limit them to Yes/No questions
- Make sure that the response options make sense
- Ask questions in a logical order
- Ask one thing at a time
- Look for people who are actually willing to share information
- Make sure that the questions are short, clear and to the point
- Define difficult terms or spell out abbreviations
- Remember that the way you interview the respondent will determine the type of information gathered, the quality of the information and how much information you can gather.



LEARN ABOUT INFORMATION GATHERING TOOLS

To learn more about information-gathering tools, have a look at this video:



<http://learn.mindset.co.za/resources/computer-applications-technology/grade-10/information-management/how-tos/01-information-sources-and-data-gathering-tools>



Saving your documents

Always save copies of everything and remember to save all your documents in a good file structure. Another thing to note is that web pages should be saved in a folder and must be named "web pages".



Activity 14.1

This activity is broken down into three sections (Sections A, B and C).

SECTION A

1. Use the instructions in the PAT and create a folder structure. Use what you have learned so far to create a folder structure.
2. Open Word.
3. Create a word-processing document.
4. Name this document: *PAT Phase1*.
5. Use the heading: *Task definition*.
6. Using the topic given below, create a task definition in the *PATPhase1* document.

Conservation is everyone's responsibility. Even if you don't spend a lot of time in nature or appreciate her beauty, everyone needs access to food, clothing, medicine, clean air and water, a variety of other resources you upon everyone's best interest to try to conserve natural systems.

Your school wants to inform learners about conservation issues, the importance of conservation and create an awareness to address these issues. The school management team wants you to investigate a conservation issue that may interest you. They also want a formal report on the investigation which they can use in their awareness campaign. The investigation should address a conservation issue/problem, the impact thereof, possible solutions and the role learners could play or any other interesting facts and statistics that will inform learners about this issue.

(Extract from question paper DBE/PAT 2014)

7. Create a sub-folder called *Task Definition* in the Phase 1 folder.
8. Save the Word document in the sub-folder.
9. Close the Word document.

SECTION B

1. In pairs, brainstorm headings that you could use for the topic. Write down a list of questions that can be applied to each heading.
2. Open the *PAT Phase1* document. Choose the headings that you think will help you complete this project. Transfer these headings and questions into the *PAT Phase 1* document.
3. Save the document.

SECTION C

1. Open another Word document.
2. Create a table. It should look similar to this:

	QUESTION	SOURCE OF INFORMATION	REASON
1.			
2.			
3.			

3. Transfer the questions that you developed in the previous section to this table.
4. Indicate the information source to answer each question.
5. Briefly explain why you think this is the best-suited information source for that specific question.
6. Save the document with the name: *Phase1Information*.
7. Save it in the correct sub-folder.

14.3 Sifting data

The information management process has three phases. This section will focus on the second phase, which is sifting the data that was received or obtained from the first phase.

INFORMATION VERSUS KNOWLEDGE

You have learned about data and information. But now, what about knowledge? How are data, information and knowledge related? It is simple: knowledge is the information that a person has collected in order to make an analysis. Let's learn more about how we will be using *knowledge* in Phase 2 of the PAT.

QUESTIONNAIRES AND SURVEYS

The PAT usually requires you to conduct a survey using a questionnaire to gather your input data. This is a crucial element, because it is here, where data is processed into meaningful information. Remember, bad data = bad information. You need the data obtained from the surveys and questionnaires to input into the spreadsheet application.

SIFTING INFORMATION

Before you can start with sifting information, you need to use information-gathering tools to find data. After you have obtained all the data from the different information sources, then only can you look at which type of data is relevant and which is not.

If you have documents in an electronic format, you can use an application, such as a word processor or PDF reader to highlight important information.

When you start with finding information for your PAT, you will probably notice that there are many websites from which you can obtain information.

However, you need to go through all the available information. From there, you need to choose the information best suited to your topic. In the PAT, the instructions given will give you an indication of what the main objective of the project or task is. The sub-headings that you choose will usually indicate what information is useful. The processing of finding this relevant information is called *sifting information*.



Activity 14.2

1. Briefly explain the difference between information and knowledge.
2. Do you think questionnaires are important in the PAT?
3. Differentiate between a questionnaire and survey.

14.4 Processing data

Data manipulation is the process where data is changed to make it easier to understand or read. The third phase of the PAT focuses on processing and using the data that is sifted in Phase 2. This section will look at how to use data manipulation, data processing and then finally, presenting the information.

KNOWLEDGE VERSUS INSIGHT/DECISION MAKING

Knowledge is the information that people collect in order to make an analysis. Decision making (wisdom) is when people put everything together and actually do something about it.

An example of decision making is when Teacher A realises that her learners are struggling with certain sections in the curriculum, especially in Term 2. This is when she needs to use this information and plan on how to remedy the situation.

THE IMPORTANCE OF INFORMATION MANIPULATION

In order to manipulate the information that you have gathered, you can do the following:

- Identifying the core meaning, which means to identify the most important points of the information that you collected.
- Summarising the information in your own words and then adding your own conclusions or opinions.
- Using graphs, tables and illustrations to show the information you collected visually.

Table 14.3: *Processing data and information*

STEP	WHAT TO DO
Group	Group data according to the questions that you asked. You can highlight the important data or information that group it accordingly.
Create hierarchies	Organise the information in an order to see where the highlighted information or data belongs.
Calculations	Using the skills that you learned in spreadsheets, you can use counting items in the spreadsheet, for example how many learners bought chips from the tuckshop on Tuesday. Functions, such as <i>MIN</i> , <i>MAX</i> and <i>AVERAGE</i> are also useful in analysing data.
Using graphs	Graphs are a good way to represent data. This will be taught when you learn about spreadsheets.

For data to be manipulated, it needs to be sorted and grouped so that the next person can understand what it means. This can be done using spreadsheets. Spreadsheets are extremely useful in sorting out data and making it look more understandable.

Microsoft Excel is an excellent spreadsheet program that allows the user to store, organise, sort and analyse information.

DATA HANDLING USING A SPREADSHEET

The results that you got from the questionnaires are a small sample of the responses that you can analyse on a spreadsheet. Unfortunately, the reality is that when it comes to data handling, people usually sit with hundreds and thousands of data items that they need to analyse.

With your PAT, however, you are only using a small sample of data to show that you are capable of processing much larger samples. At this stage, the questions that you come across when dealing with data handling would be as follows:

- How many?
- What is the most popular?
- What is the least popular?
- How many more than?
- What is the average?

PROCESSING DATA USING A SPREADSHEET

One of the best attributes of a spreadsheet program is that it can do all the calculations for you by using formulas. This application is like a calculator and can add, multiply, subtract and even divide. We will now learn about the basics of handling data using a spreadsheet.

MATHEMATICAL OPERATORS

Microsoft Excel uses the standard mathematics signs, such as the plus sign (+), minus sign (-), an asterisk to multiply (*) and a forward slash when dividing (/).

In Excel, all formulas begin with a sign.

You can also use the formulae and functions in Excel to help you process data. You will have seen most of these functions in action in your practical lessons on Excel, however, a list of basic functions is given below:

- SUM calculates the total of a range of numbers
- AVERAGE gives the average of a range of values
- MIN returns the minimum value in a list of values
- MAX returns the maximum value in a list of values
- COUNT counts the numbers in a list of values

Although these functions let you interpret large sets of data, it is not very good for visualising the data. For that, you can use charts and graphs, which will be explained in the next section.

After you have gathered the information, you need to transfer the data to an Excel spreadsheet. From there, you should use the *COUNTIF* function to count the number of answers for each question.

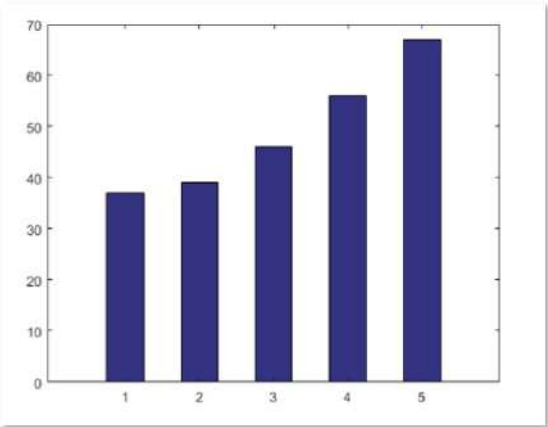
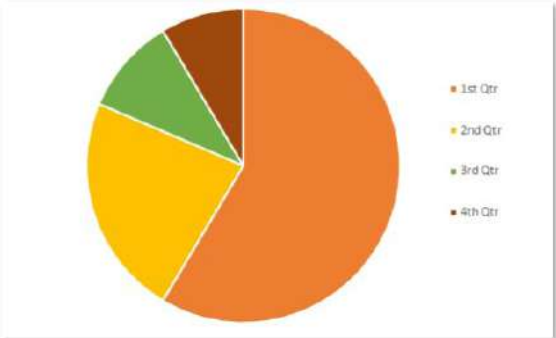
14.5 Presenting data

After you have analysed the data, you need to put it into a visual form so that the target audience can understand the results better. This is usually done in the form of graphs, tables and reports. In this section, we will look at ways in which to visualise information.

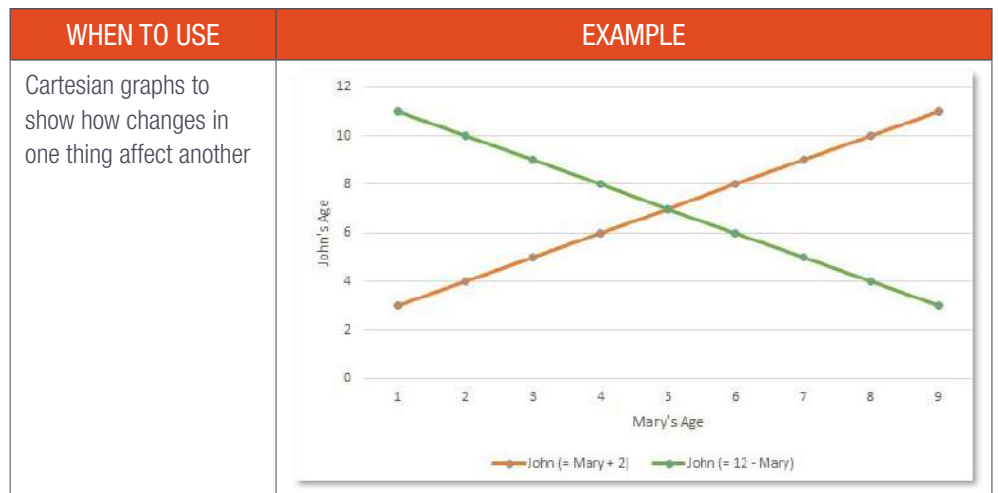
GRAPHS AND TABLES

There are different types of charts and you can choose which of them best shows the information. The line graph, bar graph and histogram, pie chart and Cartesian graphs are the four most commonly used graphs.

Table 14.4: *Application for the most common graphs*

WHEN TO USE	EXAMPLE
Bar graphs to show numbers that are independent of each other	
Pie charts when you are trying to compare the parts of a whole	
Line graphs to show how numbers have changed over time	

... continued



It is not always wise to put the graph on the same sheet as the data. It makes it cramped and might not be the easiest for a person to understand. Rather put all the graphs on the next sheet. It will make it easier for the PAT assessor to understand the information.

If you are unsure about what chart to choose, you can always go to the *Recommended Charts* tab that suggests the different types of charts based on the data.

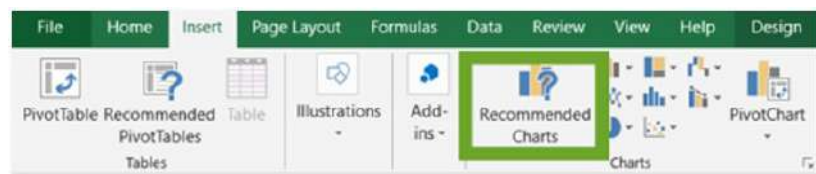


Figure 14.5: The Recommended Charts tab

REPORTS

A *report* is another way of presenting information. It is a document that contains relevant information in an organised format. It helps us to clearly state the problem, the solution that was implemented to fix the problem, as well as how effective the problem-solving solution was.

A report consists of the following sections:

- **Introduction:** The introduction is used to state the purpose of the report and provide background on the issue. It is important that you make use of your knowledge on the subject to provide a clear description of the problem that has occurred and the solution that was implemented.
- **Body:** The body is used to describe the effect of the solution that was implemented. This can include a database containing all the data regarding the solution to the problem. However, simply supplying the data is not sufficient. You also need to process and analyse the data into useful information. The information should then be presented clearly and to the point. This can be done by extracting the core meaning of the data and writing it in your own words. You can make use of graphs and tables to help simplify the information.
- **Conclusion:** This section is used to indicate whether or not the solution was a success and if any other solutions might be needed. If the solution was not successful, you need to include which steps you are planning to follow in order to make sure that the problem is resolved.

- **Bibliography/References:** It is important to always reference the information that you use in your report. It shows the evidence of where you got the information from, how recent it is, the number of sources you used and if plagiarism was committed. You need to use a fixed method to correctly cite your references.

A reference list is a list of sources that were quoted in the report. A bibliography is a list of sources that were used for the report. There are different ways to reference; however, the most common method is the Harvard referencing technique and the American Psychological Association (APA) technique.

Always remember that it is important to draft the report as soon as possible. This is to ensure that you include all the details that you have observed. The longer you wait, the better the chance of forgetting something really important.

PLAGIARISM

When writing a report, plagiarism is something that is really frowned upon. You learned about plagiarism earlier in this book.

Just to refresh your memory, plagiarism is a reproduction of someone's work as your own, without acknowledging the original source. This can be done in the form of directly using someone's findings (i.e. wording, sentences, or even the whole paragraph), pictures, etc., without referencing the original source.

When writing a report, make sure that you do not knowingly, or even unknowingly, publish or use someone else's work without acknowledging that person. To prevent plagiarism, you need to plan everything before drafting the report. Make sure you understand what the topic is about and how you are going to write the report.

It is extremely important to always hand in your own work. Therefore, when handing in your PAT, it is a good idea to hand in a declaration of authenticity, which indicates that you have not copied your work from someone else.

COPYRIGHT

Copyright, as the name suggests, is the legal right of someone to make copies of something. It is a legal right originally given to creators and they can decide who and under what conditions, a person or company may copy their invention.

INTELLECTUAL PROPERTY

Intellectual property is a term used to create something new or original. It can come in several forms. Examples could include any of the following:

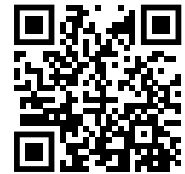
- A new invention
- Design
- Brand
- Book or journal article

Using someone else's intellectual property without their permission is considered to be a crime.



LEARNING ABOUT REFERENCING STYLES

Have a look at these videos to understand these two techniques a bit better:



Harvard <https://www.youtube.com/watch?v=6RVrhIMUaS8>



APA <https://www.youtube.com/watch?v=wIOIoMhDKc>



Something to know

The PAT guidelines give you strict instructions on what the font sizes of the main headings and body text should be. If not stipulated, however, you can use a font size of 16 pt or 18 pt for the main headings and a font size of 11 pt for the body text.

SLIDE SHOW PRESENTATION

The slide show that you need to develop for the PAT covers the same information as the report that you compiled. However, the presentation only deals with the key points and looks at the information very briefly. So, when creating a presentation, you need to think about which points are the most important.

FINALISING THE PAT

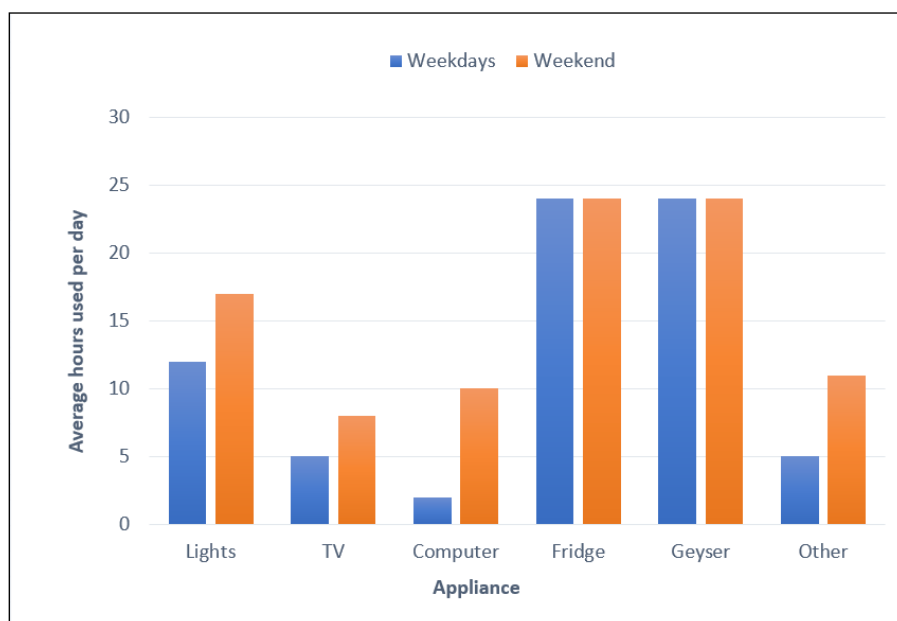
When you hand in your PAT, make sure you take note of the following:

1. Check that you have done everything that was required of you by using the marking checklist.
2. Proofread your work and make sure that your work does not have spelling and grammatical errors. If you find errors, fix them.
3. Check that all the sources have been referenced and cited properly.
4. Check the layout of the documents; how the documents look actually makes a big difference. Make sure the layout, fonts and graphs all look visually appealing.



Activity 14.3

1. List two different types of information sources.
2. Illustrate the difference between wisdom and knowledge by using an example.
3. Study the information presented in the following graph:



- a. Describe what the graph measures.
- b. Name the appliance with the least total usage.
- c. After the refrigerator and the geyser, name the appliance that is used most on week days?
- d. State why most appliances are used more during weekends than during the week.
- e. What advice will you give to households if they want to save electricity? Motivate your answer.
- f. List three different types of actions that you could perform on data to extract information, in other words, help to interpret it, or make sense of it.

REVISION ACTIVITY

1. In the 2018 PAT, Grade 12 learners were required to investigate the topic of fake news.

TOPIC 2 – FAKE NEWS

Fake news is a type of news that consists of deliberate misinformation or hoaxes spread via print and broadcast news media or online social media.

There is a huge amount of interesting discussion happening about the spread of fake news on the Internet, including how to determine if news is fake or not, and what we as individuals and companies like Facebook, Google and Twitter, among others, can possibly do to minimise the effects of it.

Your task is to:

- Investigate the current situation in **South Africa** at a national, provincial or local level
- Gather and analyse data relevant to the investigation
- Identify possible problems and solutions to any of the problems related to the topic you choose
- Identify a suitable audience (such as a school, your local community forums or business leaders) and present your research and findings in a report that would be suitable for use by the specific audience

Investigate how this topic affects the community in terms of:

- The existing problems and challenges we face in terms of this topic
- The impact in the local community, province or country
- (Partial) solutions for individuals, publishing companies, social media, et cetera

Source: <http://ecdod.co.za/documents/learners/self-study-guides/pat-cat-english>

2. Use the information above and the following outline to write a task description for the PAT.

I am doing this investigation to report to <insert target audience> about <insert PAT topic>.

I will find information on <insert PAT topic>.

I will use the following <source> <information-gathering tools> and create questionnaires to conduct interviews.

I will also use the internet to help me find more information.

After I have received all the data, I will use spreadsheets to process and analyse the data so that it converts to meaningful information.

Using this information, I will then create a report. I will also use the questionnaire answers to create a report using a word processor.

Lastly, I will translate the information into a slide presentation to show my <insert target audience>.

Replace all the phrases between angle brackets <...> with suitable words that will match the topic of fake news.

(6)

3. Thandolwethu is doing research into healthy eating habits amongst the learners in her class. She conducted a survey by asking each learner to pick their favourite vegetable from a list that she gave them. The number of votes that each vegetable received were tallied up. She then drew a graph to illustrate the votes for each vegetable.

... continued

REVISION ACTIVITY

... continued

The following table shows the results that she obtained:

VEGETABLE	NUMBER OF VOTES
Peas	1
Carrots	3
Potatoes	16
Broccoli	0
Lettuce	5
Tomatoes	12
Sweet potatoes	5
Green beans	2
TOTAL	44

- a. Use this scenario to explain the difference between data and information. (4)
- b. Thandolwethu conducted a survey to obtain her results. Explain the difference between a survey and a questionnaire. (4)
- c. What sort of graph (chart) would best illustrate the above results? (1)
- d. Use Microsoft Excel to draw a graph (chart) of Thandolwethu's results. Make sure the graph is easy to understand. You should not include the TOTAL on your graph. (4)
- e. List two pieces of information that Thandolwethu obtained from her results. (2)
4. Chloe used a questionnaire as part of her PAT. Some of the questions had "Strongly Agree", "Agree", "Disagree", "Strongly Disagree" as options for her respondents to select. She captured the responses from each person in a spreadsheet. Which spreadsheet function should she use to determine the following?
 - a. The number of respondents who answered each question (1)
 - b. The most popular response for each question based on the number of responses (1)
 - c. The number of respondents who answered "Strongly Agree" for each question (1)
5. List the four main sections of a report and state briefly what should be included in each section. (4)
6. Ntombikayise has copied and pasted three paragraphs of text taken from an online publication, into her report.
 - a. She did not acknowledge the original writer of the text in her report. What is this known as? (1)
 - b. How could she use the information without making the mistake mentioned in (a)? (2)

[TOTAL: 31]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

NO.	DID YOU ...	YES	NO
1.	Learn about the different stages in the information management process.		
2.	Learn about the problem-solving process.		
3.	Learn how to create a task definition.		
4.	Understand the different information-gathering tools and how they can be used.		
5.	Understand the connection between data, information and knowledge.		
6.	Understand the importance of using questionnaires and obtaining good data.		
7.	Learn how to use a spreadsheet application to input the data.		
8.	Learn how to sift information using a spreadsheet application.		
9.	Learn the difference between knowledge and wisdom.		
10.	Understand the importance of data manipulation.		
11.	Learn how to use data handling with a spreadsheet.		
12.	Learn how to process data with a spreadsheet.		
13.	Present data using different methods.		
14.	Understand the importance of using your own content when writing reports.		

Glossary

A

- accessory** a device added to a computer that performs an additional feature, but is not required
- active window** the currently selected window
- antivirus program** a software program designed to protect your computer against viruses
- application (app)** a type of computer software designed to perform specific tasks or a group of coordinated functions, or activities for the benefit of the user
- application software** computer software designed to perform a group of coordinated functions, tasks, or activities for the benefit of the user
- archive** an archive file is made up of one or more files that are compressed
- artificial intelligence (AI)** the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings
- authentication** a process of identifying people to make sure that they are who they say they are

B

- barcode** an optical, machine-readable, representation of data
- biometric** the technical term for body measurements and calculations
- bit** a single unit of information that can have the value of 0 or 1
- bluetooth** wireless technology that enables communication between other compatible devices
- bluetooth technology** wireless technology that works with radiowaves instead of wires or cables
- boot** the process where the computer loads the OS into its main memory or RAM and checks that all the components of the computer are in good order
- browse** reading or scanning through pages on the internet, or on your own computer
- browser** a type of software that lets you browse websites or web pages on the Web

C

- CamCard** an app used to scan, manage, synchronise and exchange business cards
- Central processing unit (CPU)** the part of a computer responsible for processing general instructions

cloud a service provided over a network that provides massive, distributed storage and processing power, and can be accessed by any internet-connected device running a web browser

cloud storage a service model in which data is maintained, managed, backed up remotely and made available to users over a network

codec a device or program that compresses data to enable faster transmission and decompresses received data

communication channel a medium through which a message is sent

communication device any device that allows a computer to connect to a network and communicate with other computers on the network

communication platform a medium through which a message is channelled to an audience

compress to make something (a file or files) smaller

computer-aided design (CAD) software used by architects, engineers, drafters, artists and others to create precision drawings or technical illustrations

computing platform a computing platform or digital platform is a set of hardware or software that enables application programs to run

container a set of files that stores information about the digital file

core a processing unit that receives instructions and performs calculations or actions based on the instructions

crowdfunding online fundraising

cursor this is an indicator that shows the current position for user interaction – for example, a cursor will show you the place where text will be inserted in a document

cursor positioning device a touchpad or trackpad is a flat surface that can detect finger contact; it is a stationary pointing device that is commonly used on laptop computers

cyclic taking place in cycles

D

data facts, figures, or information that is stored in, or used by a computer

data link a method used to connect one location to another in telecommunications to transmit and receive digital information

decompress restoring a compressed file back to its original form

default a pre-selected option adopted by a computer program

default setting the pre-existing value of a user-configurable setting that is assigned to a software application, computer program or device

digital pen a battery-operated writing instrument that allows users to digitally capture a handwritten note or drawing

digitise to convert (pictures or sound) into a digital form that can be processed by a computer

directory another name for a folder; file systems use directories to organise files within a storage device

disk drive a type of hardware that allows data to be read, written, deleted, or stored on a computer

domain the location of a website

domain name system (DNS) the phonebook of the internet

dots per inch (DPI) used when referring to a printed copy to measure the number of individual dots placed in a line within the span of one inch (2.5 cm)

downloading when your computer or smart device receives a file or data from the internet; to receive data from a remote system, typically a server such as a web server, an FTP server, an email server, or other similar systems

dynamic IP address dynamic IP addresses are not permanent; it is assigned to a node until it is connected to the network; therefore, the same node may have a different IP address every time it reconnects with the network

E

embedded device a device that contains a special-purpose computing system

emoji an electronic pictograph (picture)

encode to change something into a programming code

ethernet cable a standard network cable

F

file extension the suffix (letters) at the end of a file name that indicates what type of file it is

file path a path that defines the location of a file or folder in a computer's file system

flash disc a memory chip used for storage and transferring data, and keeps its data even if there is a break in the power supply

G

gigahertz (GHz) measuring unit to measure computer-processing speed

graphical interface element elements used by the GUI to make it easier to interact with a computer, for example checkboxes, buttons, drop lists, icons, menu bars, etc.

graphical user interface (GUI) an interface through which a user interacts with electronic devices, such as computers, hand-held devices and other appliances

graphics interchange format (GIF) a compressed image file format

H

hard disk drive (HDD) non-volatile memory hardware device that permanently stores and retrieves data from a computer

hard drive (HD) non-volatile memory hardware device that permanently stores and retrieves data from a computer

hardware refers to all the parts or components of a computer that can be installed inside, or connected to the outside of a computer

hierarchical tree a way in which data is organised into a tree-like structure

home area network (HAN) a network within a user's home that typically consists of a router and a modem

hotkey a keyboard shortcut used to perform a menu function or other common computer functions

hyperlink an icon, graphic, or text in a document that links to another file or object

hypertext mark-up language (HTML) a type of coding language used on web pages to display text, images and audiovisuals

hypertext transfer protocol (HTTP) a set of standards that allow users of the WWW to exchange information found on web pages

I

icon a picture that shows a file, folder or a program

information and communications technology (ICT) information and communications technology is a field of study related to computers and communication networks

interactive software or hardware whose behaviour changes in response to interaction with a human

internet protocol (IP) the main protocol for communication on the internet

internet service provider (ISP) a company that provides internet access to its subscribers

K

keyboarding refers to a set of skills required to operate a keyboard smoothly while typing; including being able to understand the computer keyboard layout and its functions

keylogger a software program that is used to monitor and log each key a user types on a computer keyboard

L

local area network (LAN) a network used in a small area and in close proximity to each other to share resources and information

locked documents documents with content that is difficult to change, or modify once it has been created

log-on screen the screen that allows authorised users access to the computer

M

Mac the Macintosh is a family of PCs designed, manufactured and sold by Apple Inc.

malware malicious software that is designed to change your settings, delete software, cause errors, watch browsing habits, or open computer to attacks

memory card a type of storage medium

menu a list of options or commands that the user can select to execute the proper app function

Microsoft Wordpad a free rich-text editor from Microsoft

modem a hardware device that connects a computer or router to a broadband network

N

navigation button a set of images or icons that serve as a control point to link the user to sections on a website

netiquette unwritten rules on how to use the internet

network adapter a piece of hardware that can be added to a computer, allowing it to connect to a network

network administrator someone responsible for the maintenance and operation of a network or server

network client a piece of computer hardware or software that accesses a service made available by a server

network server a computer program or device used to manage network resources

node any active, physical, electronic device attached to a network

non-volatile refers to a computer's memory that will keep data even if there is a break in the power supply

notification tray contains miniature icons for easy access to system functions, such as antivirus settings, printer, modem, sound volume, battery status and more

O

online refers to when a user, computer or device is connected to the internet

open-source software a software source code that is freely distributed

optional left to one's choice; not required or mandatory

P

password a basic security mechanism that consists of a secret phrase created using alphabetic (abc), numeric (123), alphanumeric (a14b) and symbolic characters (\$%*), or a combination thereof

personal area network (PAN) a local network designed to transmit data between PC devices and telephones

pixel a tiny dot of colour on a monitor

plagiarism taking someone else's work or ideas and passing them on as your own

platform the overall standard of a computer's hardware or software

plug-and-play (PNP) device a device that has the ability to detect and configure hardware automatically without requiring the user to physically configure the hardware with jumpers or dip switches

port a point of connection between a computer and peripheral devices, such as a mouse, keyboard, speakers, printer, monitor, flash drives and so on

prompts refers to text or symbols used to represent the system's readiness to perform the next command

proprietary relates to an owner or ownership

protocol a specific set of rules and instructions that allows computers to communicate with each other

prototype a first or preliminary version of a device from which other forms are developed

R

Random Access Memory (RAM) a very fast storage device that can only store a small amount of information at a time

read-only memory (ROM) a storage medium used with computers and other electronic devices

rich-text format (RTF) a file format standardised by Microsoft for creating formatted text files

root directory the directory at the highest level of a hierarchy

S

search engine a software program used to search for keywords on the internet to find documents or the information that was searched

shareware software that allows a user to try all, or part of the program for free before buying it

shortcut an alternative way to do something that is shorter than the one usually taken

simple mail transfer protocol (SMTP) an internet standard for sending and receiving email messages

site licence a software licence that allows a user to install a software package in several computers at the same time

SMTP server an SMTP server is the machine that takes care of the whole email delivery process

software the set of instructions that a computer uses to complete any task

solid-state drive (SSD) a storage medium that uses non-volatile memory as a means of holding and accessing data

spam describes junk email on the internet

subscriber identity module (SIM) it consists of a small chip that is used in cell phones and other mobile devices to store subscriber information

surround sound a system of stereophony involving three or more speakers surrounding the listener so as to give a more realistic effect

swiftkey a keyboard application that uses various AI technologies that enable it to predict the next word the user intends to type

switch a hardware device that filters and forwards network packets

T

taskbar the bar at the bottom of the screen that allows you to locate and launch programs through the *Start* button, or view any program that is currently open

taxonomy the science of naming things; in this case, email addresses

tiles a new feature consisting of icons on the Windows *START* screen introduced by Microsoft

touch screen a computer display screen that is sensitive to pressure; a user interacts with the computer by touching pictures or words on the screen

touch typing typing without looking at the keys

Trojan horse a program that appears to be safe, but is performing tasks, such as giving access to your computer or sending personal information to other computers

U

ultra-portable very slim, lightweight and easy to carry around

uniform resource locator (URL) a standard naming convention for addressing documents that are accessible over the internet

universal serial bus (USB) the most common type of computer port used in today's computers

uploading when your computer or smart device sends a file or data to somewhere on the internet

username a word, phrase, number or combination of characters that uniquely identify a user on any computing device, or related service that requires user authentication

V

virulent hostile or harmful

virus a program designed to cause damage, steal personal information, modify data, send emails, display messages, or a combination of these

volatile memory computer storage that only maintains its data while the device is powered

W

web refers to the world wide web (WWW), which is a graphical interface for the internet

web browser a software program to present and explore content on the WWW

web server a computer or collection of computers used to deliver web pages and other content to multiple users

webmail a service provided by companies and an ISP to enable you to access your email over an internet page instead of an email client

web page a document commonly written in HTML that is accessible through the internet, or other network, using an internet browser

website a collection of web pages

wide area network (WAN) a collection of computers and network resources connected via a network over a geographic area

Wi-Fi a technology that uses radiowaves to provide network connectivity; established using a wireless adapter to create hotspots – areas in the vicinity of a wireless router that are connected to the network and allow users to access internet services

Wi-Fi hotspot a location that enables users to connect to a network or the internet with their wireless adapters

window an area on the screen that displays information for a specific program

wireless telecommunications in which electromagnetic waves (rather than some form of wire) carry the signal over part, or all of the communication path

wireless access point a wireless receiver that enables a user to connect wirelessly to a network or the internet

wireless local area network (WLAN) a type of local network that uses radiowaves to transmit data wireless technology a technology that uses radiowaves to send and receive data